

JAVA DEVELOPER INTERNSHIP

TASK 1

JAVA BASICS & OOP FUNDAMENTALS

Internship Organization

ALFIDO TECH

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GitHub Repository: [vermashreya127-blip/Task1-Java-Basics](https://github.com/vermashreya127-blip/Task1-Java-Basics): Java Basics and OOP Fundamentals internship task using Java.

Objective

The objective of this task is to understand the fundamentals of Java programming and Object-Oriented Programming (OOP). This project demonstrates the use of loops, conditional statements, arrays, classes, objects, inheritance, and polymorphism through practical Java programs.

Tools Used

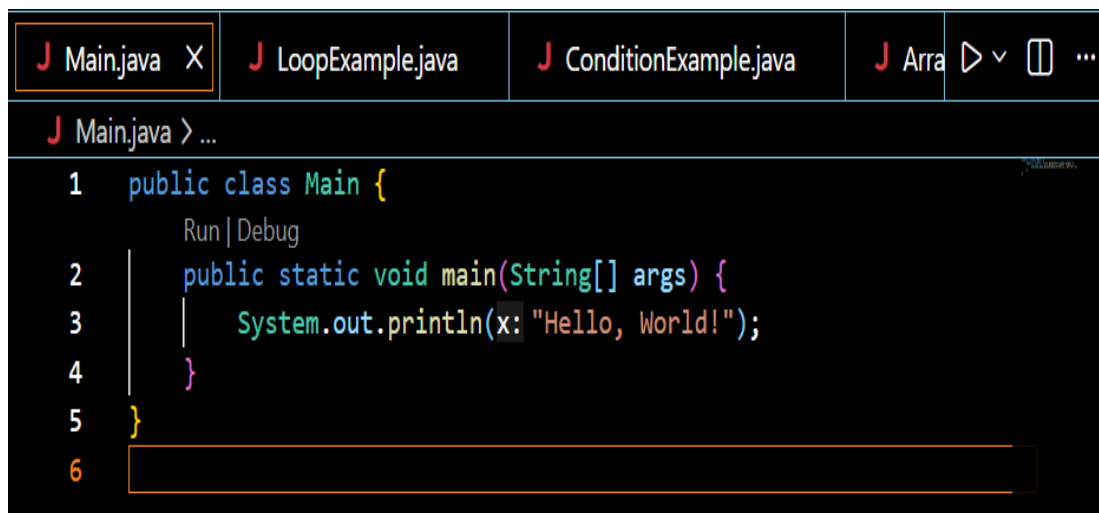
- Java JDK 21
- Visual Studio Code (VS Code)
- GitHub

Program1: Hello World

Description:

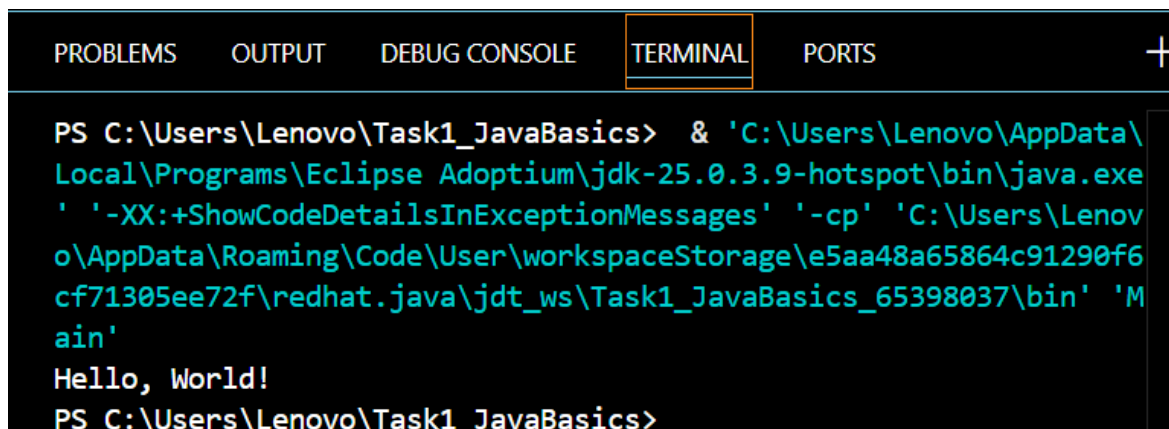
This program prints "Hello, World!" on the screen. It is used to verify that Java is installed and working correctly.

Code:



```
J Main.java X J LoopExample.java J ConditionExample.java J Arra ▶ ▢ ...
J Main.java > ...
1 public class Main {
    Run | Debug
2     public static void main(String[] args) {
3         System.out.println(x: "Hello, World!");
4     }
5 }
6
```

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS +
PS C:\Users\Lenovo\Task1_JavaBasics> & 'C:\Users\Lenovo\AppData\Local\Programs\Eclipse Adoptium\jdk-25.0.3.9-hotspot\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Lenovo\AppData\Roaming\Code\User\workspaceStorage\e5aa48a65864c91290f6cf71305ee72f\redhat.java\jdt_ws\Task1_JavaBasics_65398037\bin' 'Main'
Hello, World!
PS C:\Users\Lenovo\Task1_JavaBasics>
```

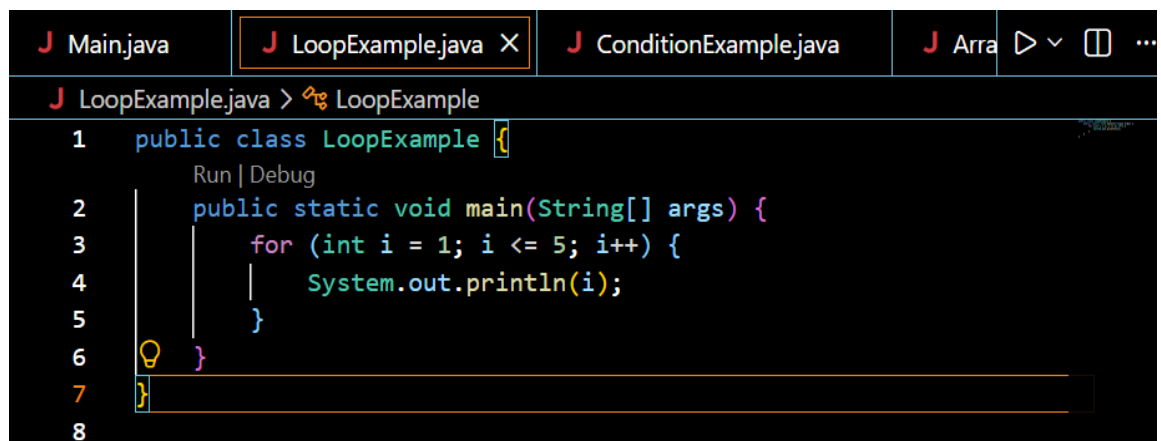
Program2:

Loop Example

Description:

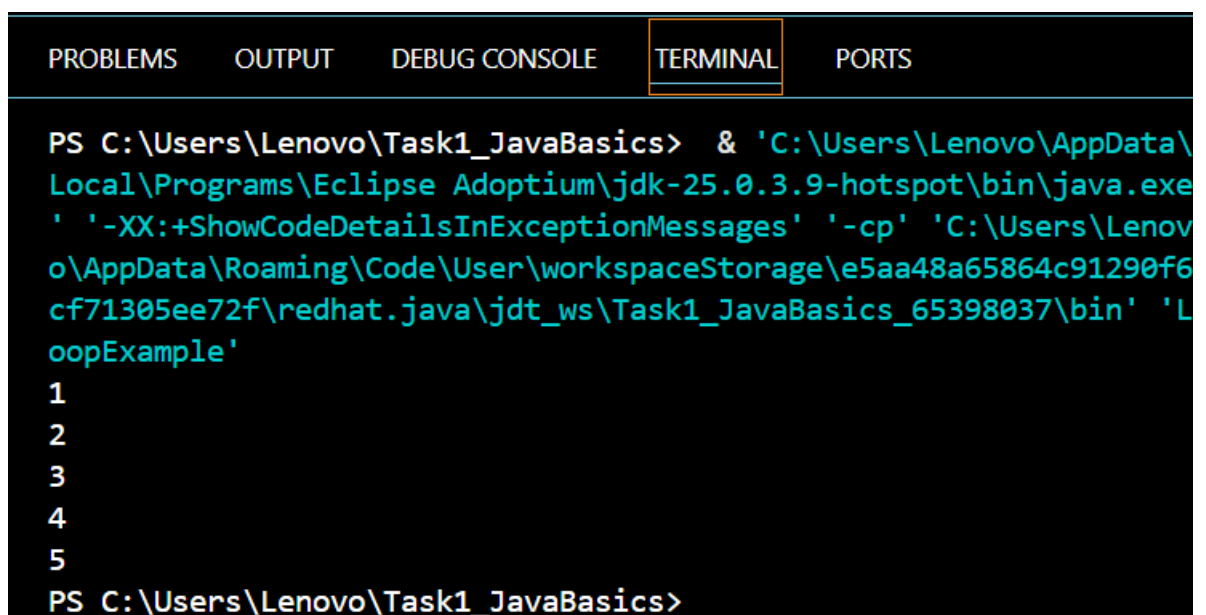
This program uses a for loop to print numbers from 1 to 5.

Code:



```
J Main.java J LoopExample.java X J ConditionExample.java J Arra ▶ ▼ □ ...
J LoopExample.java > LoopExample
1 public class LoopExample {
    Run | Debug
2     public static void main(String[] args) {
3         for (int i = 1; i <= 5; i++) {
4             System.out.println(i);
5         }
6     }
7 }
8
```

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\Lenovo\Task1_JavaBasics> & 'C:\Users\Lenovo\AppData\Local\Programs\Eclipse Adoptium\jdk-25.0.3.9-hotspot\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Lenovo\AppData\Roaming\Code\User\workspaceStorage\e5aa48a65864c91290f6cf71305ee72f\redhat.java\jdt_ws\Task1_JavaBasics_65398037\bin' 'LoopExample'
1
2
3
4
5
PS C:\Users\Lenovo\Task1_JavaBasics>
```

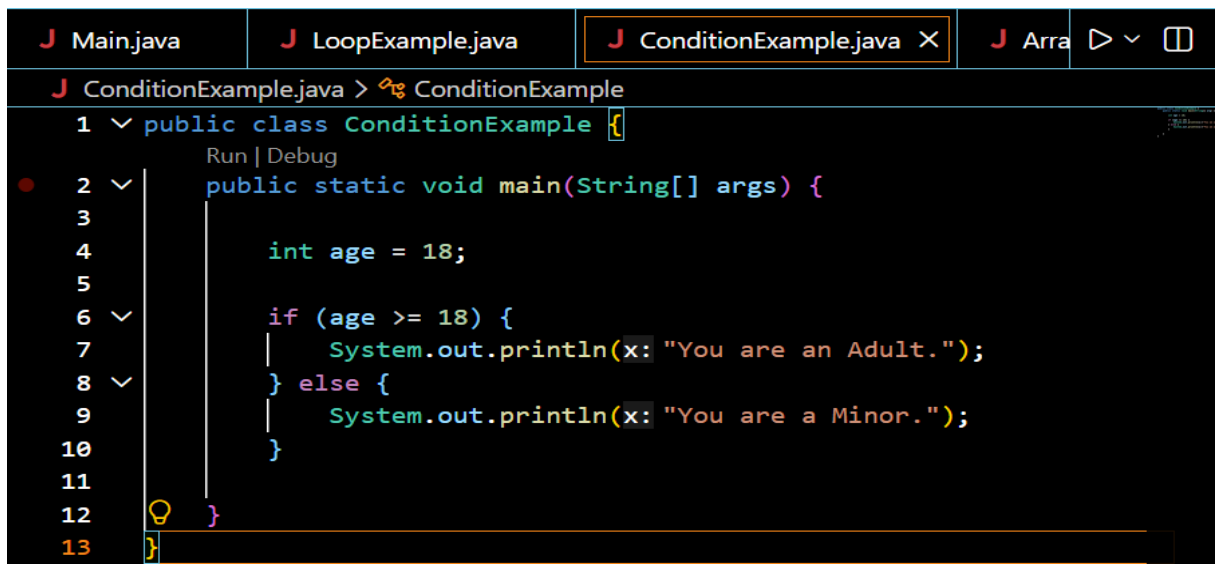
Program3:

Condition Example

Description:

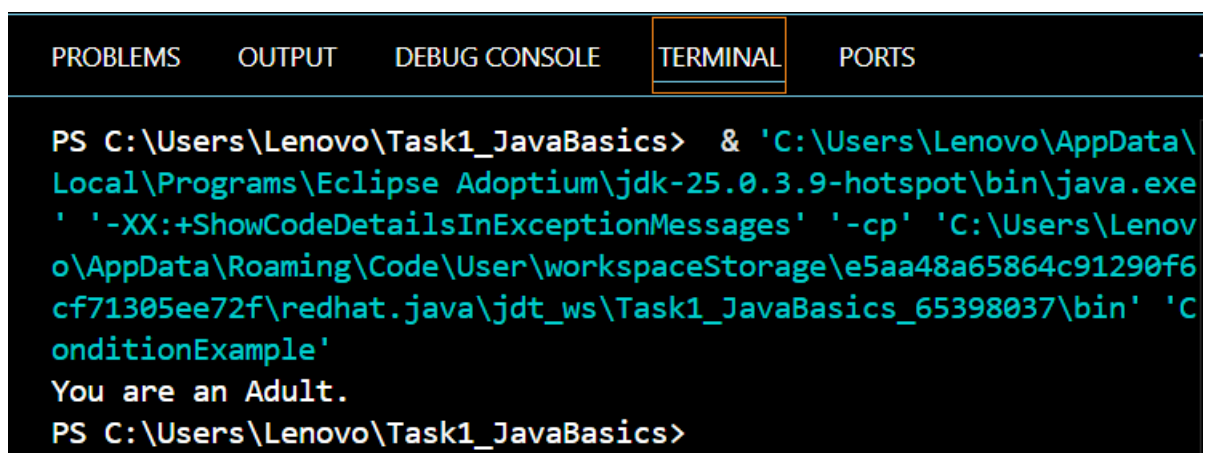
This program demonstrates the use of an if-else statement to check a condition and display the appropriate output.

Code:



```
J Main.java J LoopExample.java J ConditionExample.java X J Arra ▶ ▼ □
J ConditionExample.java > ⚙ ConditionExample
1  public class ConditionExample {
    Run | Debug
2  public static void main(String[] args) {
3
4      int age = 18;
5
6      if (age >= 18) {
7          System.out.println(x: "You are an Adult.");
8      } else {
9          System.out.println(x: "You are a Minor.");
10     }
11
12 }
13 }
```

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\Lenovo\Task1_JavaBasics> & 'C:\Users\Lenovo\AppData\Local\Programs\Eclipse Adoptium\jdk-25.0.3.9-hotspot\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Lenovo\AppData\Roaming\Code\User\workspaceStorage\e5aa48a65864c91290f6cf71305ee72f\redhat.java\jdt_ws\Task1_JavaBasics_65398037\bin' 'ConditionExample'
You are an Adult.
PS C:\Users\Lenovo\Task1_JavaBasics>
```

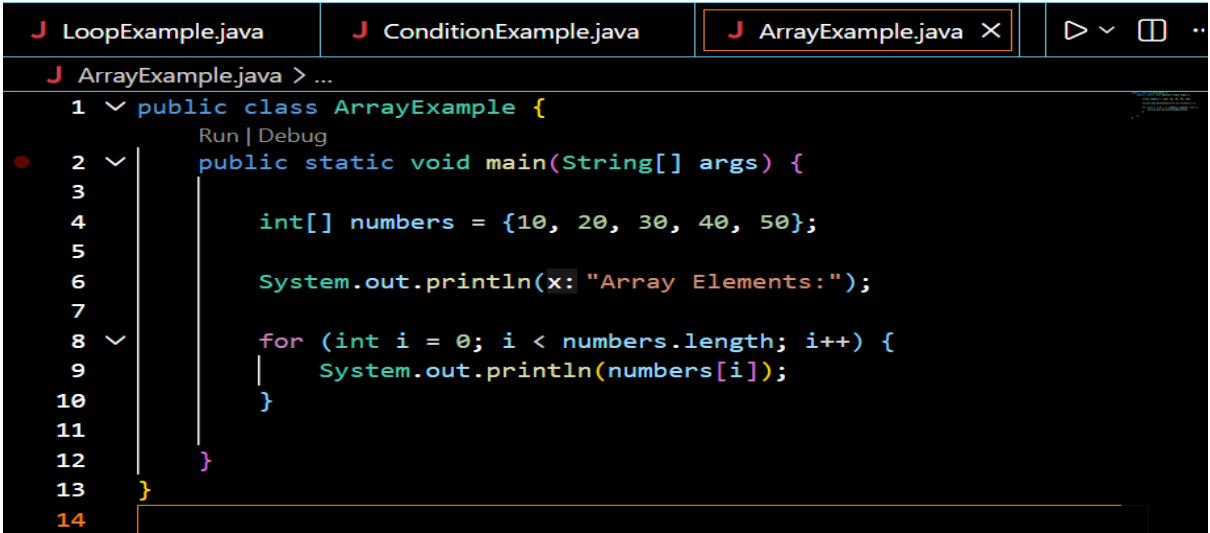
Program4:

Array Example

Description:

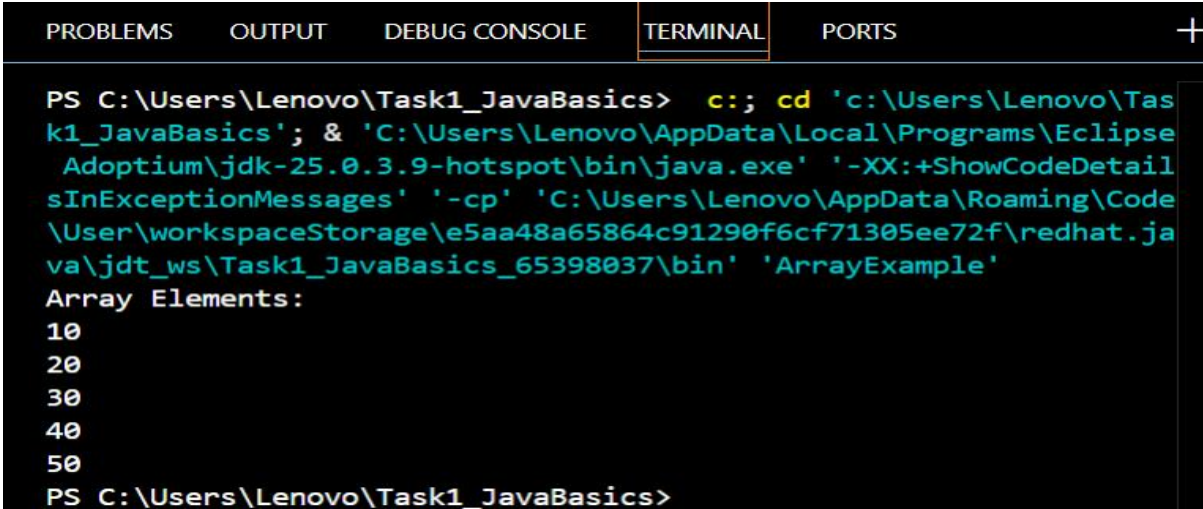
This program demonstrates the use of arrays in Java and prints all array elements using a loop.

Code:



```
1 public class ArrayExample {
2     public static void main(String[] args) {
3
4         int[] numbers = {10, 20, 30, 40, 50};
5
6         System.out.println("Array Elements:");
7
8         for (int i = 0; i < numbers.length; i++) {
9             System.out.println(numbers[i]);
10        }
11    }
12 }
13
14
```

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\Lenovo\Task1_JavaBasics> c:; cd 'c:\Users\Lenovo\Task1_JavaBasics'; & 'C:\Users\Lenovo\AppData\Local\Programs\Eclipse Adoptium\jdk-25.0.3.9-hotspot\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Lenovo\AppData\Roaming\Code\User\workspaceStorage\e5aa48a65864c91290f6cf71305ee72f\redhat.java\jdt_ws\Task1_JavaBasics_65398037\bin' 'ArrayExample'
Array Elements:
10
20
30
40
50
PS C:\Users\Lenovo\Task1_JavaBasics>
```

Program5:

Student (Class & Object)

Description:

This program demonstrates the concepts of class, object, and methods in Java.

Code:

```
1 class Student {
2     String name;
3     int age;
4
5     void display() {
6         System.out.println("Name: " + name);
7         System.out.println("Age: " + age);
8     }
9
10    Run | Debug
11    public static void main(String[] args) {
12
13        Student s1 = new Student();
14
15        s1.name = "Shreya";
16        s1.age = 20;
17
18        s1.display();
19    }
```

Output:

```
Local\Programs\Eclipse Adoptium\jdk-25.0.3.9-hotspot\bin\java.exe
'-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Lenovo\AppData\Roaming\Code\User\workspaceStorage\e5aa48a65864c91290f6cf71305ee72f\redhat.java\jdt_ws\Task1_JavaBasics_65398037\bin' 'Student'
Name: Shreya
Age: 20
PS C:\Users\Lenovo\Task1 JavaBasics>
```

Program6:

AnimalTest (Inheritance & Polymorphism)

Description:

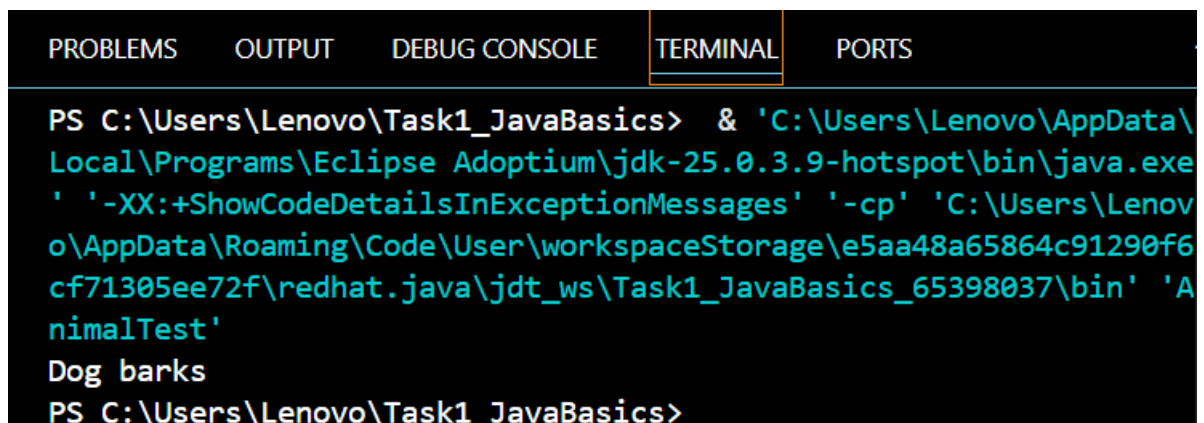
This program demonstrates inheritance and polymorphism using Java classes.

Code:



```
1 class Animal {
2     ...
3 }
4
5
6
7 class Dog extends Animal {
8     @Override
9     void sound() {
10        System.out.println(x: "Dog barks");
11    }
12 }
13
14 public class AnimalTest {
15     Run | Debug
16     public static void main(String[] args) {
17         Animal a = new Dog();
18         a.sound();
19     }
20 }
```

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\Lenovo\Task1_JavaBasics> & 'C:\Users\Lenovo\AppData\Local\Programs\Eclipse Adoptium\jdk-25.0.3.9-hotspot\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Lenovo\AppData\Roaming\Code\User\workspaceStorage\e5aa48a65864c91290f6cf71305ee72f\redhat.java\jdt_ws\Task1_JavaBasics_65398037\bin' 'AnimalTest'
Dog barks
PS C:\Users\Lenovo\Task1_JavaBasics>
```

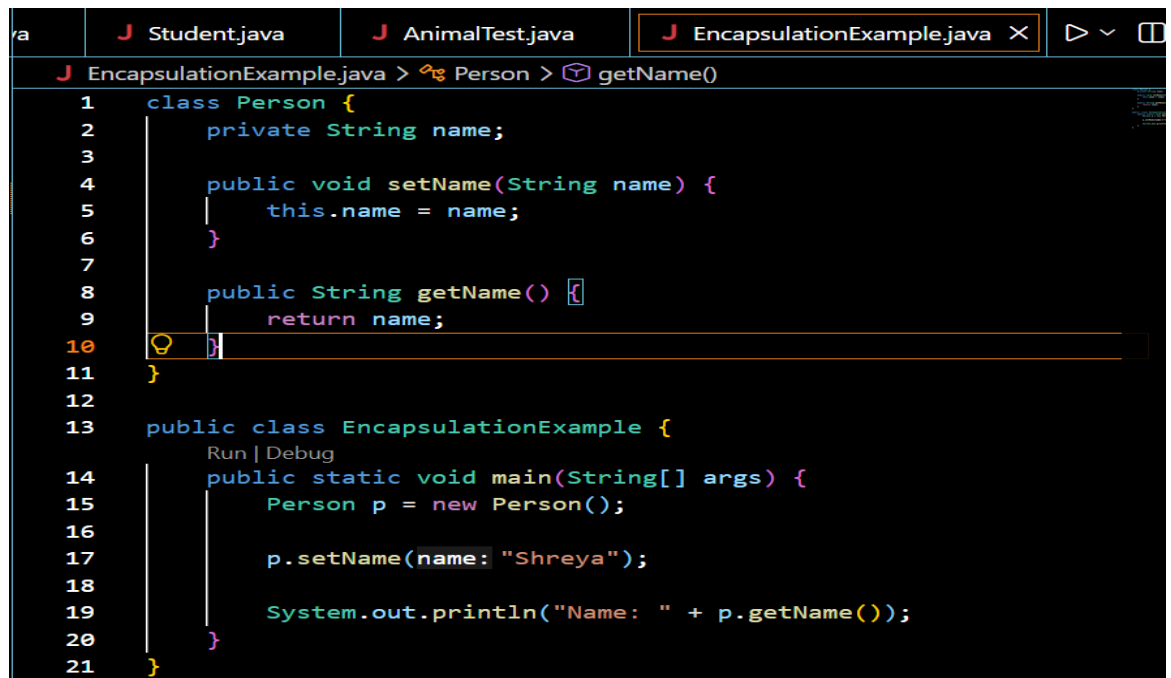

Program7:

Encapsulation Example

Description:

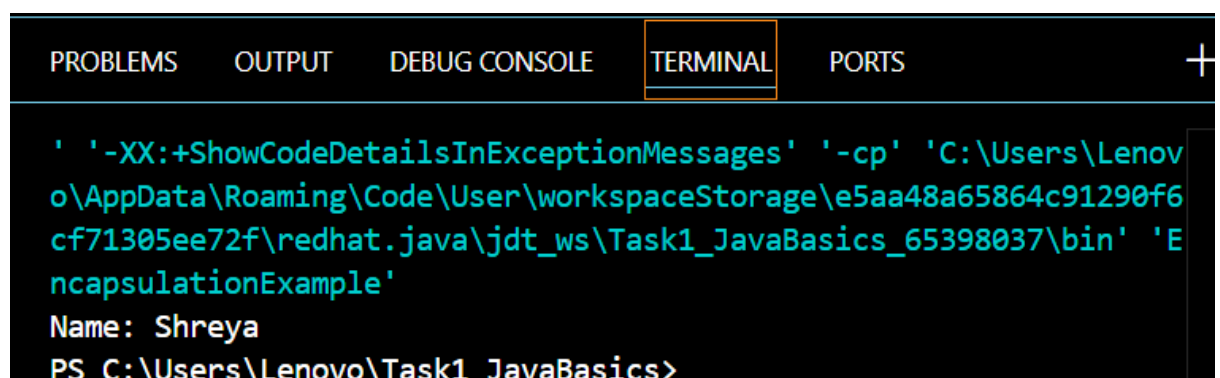
This program demonstrates encapsulation in Java by declaring data members as private and accessing them through getter and setter methods.

Code:



```
1 class Person {
2     private String name;
3
4     public void setName(String name) {
5         this.name = name;
6     }
7
8     public String getName() {
9         return name;
10    }
11 }
12
13 public class EncapsulationExample {
14     Run | Debug
15     public static void main(String[] args) {
16         Person p = new Person();
17         p.setName(name: "Shreya");
18
19         System.out.println("Name: " + p.getName());
20     }
21 }
```

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS +
'-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Lenovo\AppData\Roaming\Code\User\workspaceStorage\e5aa48a65864c91290f6cf71305ee72f\redhat.java\jdt_ws\Task1_JavaBasics_65398037\bin' 'EncapsulationExample'
Name: Shreya
PS C:\Users\Lenovo\Task1 JavaBasics>
```

OOP Concepts

Class

A class is a blueprint or template used to create objects. It defines the properties (variables) and behaviors (methods) that objects of that class will have.

Object

An object is an instance of a class. Objects allow us to access the properties and methods defined within a class.

Inheritance

Inheritance is an OOP concept that allows one class to acquire the properties and methods of another class. It promotes code reusability and reduces redundancy.

Polymorphism

Polymorphism allows a method to perform different actions based on the object that invokes it. It increases flexibility and improves code maintainability.

Encapsulation

Encapsulation is the process of binding data and methods together within a class and restricting direct access to data using access modifiers. It helps protect data and maintain security.

Learning Outcomes

- Learned Java fundamentals.
- Understood loops, conditions, and arrays.
- Implemented OOP concepts.
- Improved problem-solving skills.

Conclusion

This task helped me understand Java programming fundamentals and Object-Oriented Programming concepts through practical implementation. It strengthened my coding skills and provided hands-on experience with Java development.